

NVIDIA Unreal Engine DLSS plugins

8.3.0 September 2025

Component	Note
Unreal Engine	5.6 to 5.4 deprecated: 5.3, 5.2
NVIDIA RTX	5.6
UE Platforms	Win64, WinGDK
Streamline SDK	2.9.0
NGX SDK	310.4.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
 - DLSS Super Resolution (DLSS-SR)
 - Deep Learning Anti Aliasing (DLAA)
 - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
 - RTX Dynamic Vibrance
 - Reflex Low Latency
- D3D12RHI
 - DLSS Ray Reconstruction (DLSS-RR)
 - DLSS Frame Generation (DLSS-FG)
 - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

Note: Support for deprecated UE versions will be removed in the future, but not before the next UE 5.X version release

New Features

- Updated to SL SDK 2.89.0
- Updated to NGX SDK 310.4.0
- Implement auto mode for modular DLSS upscaler

General Changes

- Deprecated support for UE 5.3, UE 5.2
- Added 'Getting Started with NVIDIA DLSS 4 for Unreal Engine.pdf' back, which was unintentionally missing from previous releases.
- Downgrade some Streamline SDK log messages to Verbose and VeryVerbose as they get routed to `LogStreamlineAPI`
- Use `RDG_EVENT_SCOPE_STAT` instead of deprecated `RDG_GPU_STAT_SCOPE` in version 5.6
- Added various functional tests
- Added *Incompatibilities with the Mobile Forward Renderer* sections to the plugin documentations

Breaking Changes

- Removal of DLSS-SR/RR sharpness
- Removed previously deprecated DLSS-SR and DLSS-RR presets
 - Projects which used those will be changed to use the respective default preset
- Fixed DLSS-SR not respecting alpha output by default. May cause a slight performance difference when alpha output is enabled. Set `r.NGX.DLSS.EnableAlphaUpscaling 0` to restore old default behavior.

DLSS Super Resolution (DLSS-SR)

- Removed DLSS-SR presets A,B,C,D,E
- Removal of DLSS sharpness,
 - applications are encouraged to use NIS for sharpening, as outlined in the RTX UI Guidelines

- Removed CVars
 - `r.NGX.DLSS.Sharpness`
 - `r.NGX.DLSS.PreferNISSharpen`
 - Removed Blueprint `UDLSSLibrary::SetDLSSSharpness` and `UDLSSLibrary::GetDLSSSharpness`
- deprecated `UDLSSLibrary::GetDLSSModeInformation::OptimalSharpness` output so future integrations don't connect to it
 - The now deprecated `OptimalSharpness` always returns 0.35, which is what the underlying `NGXGetOptimalSettings` has been returning.
- Fixed DLSS-SR not respecting alpha output by default. May cause a slight performance difference when alpha output is enabled. Set `r.NGX.DLSS.EnableAlphaUpscaling 0` to restore old default behavior.
 - CVar functionality change
 - `r.NGX.DLSS.EnableAlphaUpscaling` is now based off `r.PostProcessing.PropagateAlpha` by default (-1).
- Removed `r.NGX.DLSS.AutomationTesting`. The feature did not work as advertised. If you need to disable the NGX library from being loaded, use the `-ngxdisable` command line option or the `r.NGX.Enable` console variable.
- Added `FDLSSModule::GetDLSSDenoiser()`. Use this to work around other poorly behaved denoiser plugins that don't pass control down the denoiser chain when they're not active.

DLSS Ray Reconstruction (DLSS-RR)

- See DLSS-SR
- Removed DLSS-RR presets A,B,C
- On engine versions 5.4 and later, Lumen bilateral filtering is now automatically disabled while DLSS-RR is enabled in the blueprint library/methods.

DLSS Frame Generation (DLSS-FG)

Bug Fixes

- UE-1345 RenderGraphValidation Issue involving resource `CustomDepth` and the pass `BiasCurrentColorPass`
- 5416920 - [DLSS4 Test App] - 4k resolution option does not function correctly
- 5416658 - Editor crashes during second sequence of the Movie Render Queue (UE 5.2)

Known Issues

General:

- New: 5504512 - Heavily amplified instability observed with DLSS-SR Ultra Performance option (UE 5.6 & 5.5)
- 5417010 - Stand-alone game crashes on launch (UE 5.6)
- 5416670 - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (5.5, 5.6)
- 5356855: Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- 5416522 - Unreal Engine Editor crashes when launched with Vulkan API (5.2)
- 5417003 - Editor crashes after MovieRenderQueue ends (5.6)
- 5417021 - Editor crashes after toggling through different DLSS-RR presets while playing the project in a new window (5.2)
- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display
- 4202194 - Unchecking "Warn about incompatible plugin tools" on VK does not remove the dialog box, even after restarting the editor

DLSS4Test App Specific:

- New: 5502683 - [Vulkan][DLSS4 Test App] - DLSS4Test-Win64-Shipping crashes on launch with non-NV GPUs (UE 5.6)
- New: 5502626 - [Vulkan][DLSS4 Test App] - DLSS4Test application exhibits heavy corruption on non-NV GPUs (UE 5.6)
- New: 5502713 - [DLSS4 Test App] - Application crash observed upon cycling through 'constrained view' and 'View Rect Offset' options with DLSS-SR - performance modes (UE 5.6)
- New: 5417032 - [DX12][DLSS4 Test App] - Application launches to blank window on non-NV GPUs (UE 5.5)
- New: 5228046 - [SDK/Plugin Release][DLSS4Test app][RTX] Image corruption occurs when View Rect Offset is adjusted
- New: 5416522 [Vulkan] - Unreal Engine Editor crashes when launched with Vulkan API (UE 5.2)
- New: 5502632 - [DLSS4Test-Win64-Shipping] Single view is the only functional Multi-Viewport option (UE 5.2)
- New: 5504701- [DLSS4Test App] Verification HUD is showing "Enabled" for Reflex even after disabling it (UE 5.2)
- New: 5502698 – [DLSS4Test App] Corruption observed when changing "View Rect Offset" slider from 1 to 2 with RX enabled (UE 5.2)

- 5416464 - Application exhibits heavy corruption
- 5416489 - Application crashes upon changing DLSS-SR modes (UE 5.2)
- 5416646 - DLSS-SR is not engaged on launch despite 'auto' mode being selected
- 5416702 - Heavy stutter observed when changing constrained views with DLSS-FG enabled (5.5)
- 5416990 - Only the DLSS4 Log populates when logging
- 5417032 - Application launches to blank window on non-NV GPUs (5.5, 5.6)
- 5228046 Image corruption occurs when View Rect Offset is adjusted
- 5046562 New Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 Amplified instability observed on thin lines with different constrained views
- 5046507 Amplified noise on reflections with DLSS-SR Enabled
- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

8.2.0 July 2025

Component	Note
Unreal Engine	5.2 to 5.6
NVIDIA RTX	5.6
UE Platforms	Win64, WinGDK
Streamline SDK	2.8.0
NGX SDK	310.3.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
 - DLSS Super Resolution (DLSS-SR)
 - Deep Learning Anti Aliasing (DLAA)
 - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
 - RTX Dynamic Vibrance
 - Reflex Low Latency
- D3D12RHI
 - DLSS Ray Reconstruction (DLSS-RR)
 - DLSS Frame Generation (DLSS-FG)
 - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

New Features

- Updated to SL SDK 2.8.0
- Updated to NGX SDK 310.3.0
- Support the IUpscalerModularFeature interface added in Unreal Engine 5.6 to allow other features to dynamically discover DLSS support when available.
 - Automatically adds DLSS Super Resolution/DLAA as an nDisplay plugin upscaling option with Unreal Engine 5.6 or later.

General Changes

- Deprecated DLSS-SR presets A,B,C,D,E
- Use the new slSetTag API with frame ids

- Uses newer Streamline SDK API to tag resources for various frames in flight
- Fallback to old behavior can be used with `Use slSetTag` (deprecated) plugin and plugin override setting as well as `-sl{no}settag` command line arguments
- Add plugin side support for upcoming NVRTX DLSS-RR enhancements
- StreamlineLogSink, wrap implementation in `#if !NO_LOGGING #endif` sandwich
- FStreamlineRHI::OnSwapchainCreated and FStreamlineRHI::OnSwapchainDestroyed, change Log verbosity to Verbose
- Add `-sl{no}ota` command line option to enable/disable OTA in addition to the project settings
 - Added plugin user override setting
- Restructured and reorganized the DLSS-SR and DLSS-Project settings
- Updated DLSS-FG documentation about possible IQ issues with select engine features and plugins
- Updated DLSS-SR documentation about the difference between "engine needs run in TAA mode" vs the RTX UI guidelines suggesting to "disable temporal AA"

Breaking Changes

- `UDLSS::Auto` mode returned by `UDLSSLibrary` BP library functions that haven't done so before. This means existing BPs that manually add `UDLSS::Auto` to UI elements might need adjustments to remove duplicate Auto modes in the UI

DLSS Super Resolution (DLSS-SR)

- Deprecated DLSS-SR presets A,B,C,D,E
- `UDLSSLibrary::GetSupportedDLSSModes` returns `UDLSSMode::Auto`
- `UDLSSLibrary::IsDLSSModeSupported(UDLSSMode::Auto)` returns `true`
- added note to *Enabling DLSS (Engine Side)* section about engine temporal AA mode vs RTX UI Guidelines "disable temporal AA mode"
- group DLSS-SR presets into separate DLSS-SR settings group

DLSS Ray Reconstruction (DLSS-RR)

- UE-1251 Add SSS and DoF guide buffers for DLSS-RR
- `UDLSSLibrary::GetSupportedDLSSModes` returns `UDLSSMode::Auto`
- `UDLSSLibrary::IsDLSSModeSupported(UDLSSMode::Auto)` returns `true`
- show paths of DLSS-RR binaries in the project settings to match those for DLSS-SR
- group DLSS-RR presets into separate DLSS-SR settings group

DLSS Frame Generation (DLSS-FG)

- Added *HDR and UI Color and Alpha* section to DLSS-FG documentation
- Added *Possible Reduced IQ with OpenColorUI plugin* section to DLSS-FG documentation

Bug Fixes

- UE-1222 Cvar `r.Streamline.InitializePlugin` declared too late.
- UE-1149 address clang compatibility and other issues reported by partners
- UE-1152 Fixes for const correctness and logging reported by partners
- 5158150 Enabling 'Load Debug Overlay' from the project settings causes a D3D device failure / GPU Crash with 'Standalone Game' mode

Known Issues

General:

- New: 5417010 - Stand-alone game crashes on launch (UE 5.6)
- New: 5416670 - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (5.5, 5.6)
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- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

8.1.0 July 2025

Note: New UE plugin version scheme to separate it from the DLSS 4 product version scheme.

Component	Note
Unreal Engine	5.2 to 5.6
NVIDIA RTX	5.6
UE Platforms	Win64, WinGDK
Streamline SDK	2.7.30
NGX SDK	310.2.0

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- D3D12RHI, D3D11RHI, VulkanRHI
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 - RTX Dynamic Vibrance
 - Reflex Low Latency
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Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

New Features

- Support for UE 5.6
 - **Note:** DLSS plugin side changes to support nDisplay are not this release and will be added in the future
- Improved -d3ddebug and -rvalidation compatibility for all supported UE versions
- Expose the "bias current color mask" DLSS-SR/RR inputs to allow UE applications to improve feature tracking

General Changes

- UE 5.6 support
 - UE 5.6 reworked the resource state handling between the D3D12RHI and RDG. The D3D12RHI resource state tracker got removed, and resource states are tracked by the RDG now. This required changes to the UE plugins to follow suit
 - UE 5.6 added new engine methods to allow the UE DLSS & Streamline plugin input and output resources to be made resident in memory. This avoids GPU access violations when the system is VRAM oversubscribed.
- Changes to UE 5.5 and older to improve to debug layer and memory residency compatibility, guided by the 5.6 changes
- Added *Graphics Debug and Validation Layers* sections to the Streamline and DLSS-SR/RR plugin documentation

- Consolidated various sections into new *Plugins and Platforms* section in the documentation
 - Added *Plugins and other Platforms* and *Plugins and Custom Engines* subsections
- Expose DLSS-SR/RR "bias current color mask" input to UE applications

DLSS Super Resolution (DLSS-SR)

DLSS may “erase” a particular feature or can produce a “ghost” or trail behind a moving feature. If a problematic asset is discovered during testing, one option is to instruct DLSS to bias the incoming color buffer over the colors from previous frames.

See *Enabling Bias Current Color Mask for DLSS* section in the DLSS-SR/RR UE plugin documentation for UE specifics.

DLSS Ray Reconstruction (DLSS-RR)

The DLSS-SR "bias current color mask" section also applies to DLSS-RR

Bug Fixes

- UE-1164 UE SL Reflex Plugin - Time spent in `slReflexSleep` is not treated as idle time, so the measured game thread timings will be incorrect when GPU limited with Reflex enabled.
- UE-1144 UE 5.5 Fix GPU memory access due to DLSS output texture not being resident
- 5048577 Application crash due to Error: Render subrect base and size exceed the dimensions of the input Color image
- 5228061 'Auto' Mode in the DLSS4Test app doesn't automatically adjust DLSS resolution, and only inherits the last set DLSS resolution

Known Issues

General:

- New: 5356855: Running the packaged sample in vulkan mode causes significant render instability or an immediate crash
- 5158150 Enabling 'Load Debug Overlay' from the project settings causes a D3D device failure / GPU Crash with 'Standalone Game' mode

Editor Specific:

- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display

DLSS4Test App Specific:

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4.0.2-2.7.30 April 2025

Component	Note
Unreal Engine	5.2 to 5.5
NVIDIA RTX	5.4
UE Platforms	Win64, WinGDK
Streamline SDK	2.7.30
NGX SDK	310.2.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
 - DLSS Super Resolution (DLSS-SR)
 - DLSS Ray Reconstruction (DLSS-RR)
 - Deep Learning Anti Aliasing (DLAA)
 - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
 - RTX Dynamic Vibrance
 - Reflex Low Latency
- D3D12RHI
 - DLSS Frame Generation (DLSS-FG)
 - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

New Features

DLSS Super Resolution

- Added a new default preset, Preset K (Transformer model)

DLSS Ray Reconstruction

- Added a new preset, Preset E (Transformer model)
- Deprecated DLSS-RR presets A,B,C

General Changes

Application developer facing:

- split the DLSS-RR presets off the DLSS-SR preset so they can be changed independently via cvars and project settings in the editor
- added support to detect deprecated DLSS-SR/RR presets and set those to default and then show an editor message box & log message.
- add tooltips to the DLSS-SR/RR presets based on the descriptions in the SDK docs & headers

Plugin internal:

- don't create empty NGX log folders
- removed `r.NGX.EnableOtherLoggingSinks`

DLSS Frame Generation (DLSS-FG)

- added cvar for `eRetainResourcesIfOff` flag that can be used to reduce stutter when changing DLSS-FG settings in a UI

DLSS Super Resolution (DLSS-SR)

A new Preset K project setting has been added to select the DLSS-SR Transformer model

DLSS Ray Reconstruction (DLSS-RR)

DLSS Ray Reconstruction supports stock UE to a certain degree. However the engine modifications in the NVRTX 5.4 branch additional input buffers which will greatly improve the overall quality of DLSS-RR.

A new Preset E project setting has been added to select the DLSS-RR Transformer model

Presets A,B,C have been deprecated and been hidden from the project settings. They can still be selected via the `r.NGX.DLSSRR.Preset` cvar but it is not recommended to do so.

The experimental Substrate material system is not compatible with DLSS-RR. DLSS-RR uses the traditional G-Buffer which is used to compute guide buffers such as diffuse and specular albedo. Substrate however does not provide a traditional guide Buffer. This will be addressed in a future release.

Bug Fixes

- UE-1111 DLSS-SR Preset J cannot be selected due to type casting but
- UE-1119 Crash in StreamlineCore when the deprecated Streamline plugin is not present
- UE-1122 In shutdown NGX check for StreamlineCore instead of Streamline plugin
- UE-1124 StreamlineLatewarp: fix 'bDrawInGame': is not a member of 'FPrimitiveSceneInfo'
- UE-1113 Ensure StreamlineReflex detects its own tick rate handler
- UE-1132 UStreamlineLibraryReflex::GetReflexMode() return Invalid enum when mode is EStreamlineReflexMode::Boost
- UE-1150 FTexture2DRHIRef deprecation warnings for the NIS plugin
- UE-1110 Fix FTexture2DRHIRef deprecation warnings StreamlineLatewarp

Known Issues

General:

- New: 5158150 Enabling 'Load Debug Overlay' from the project settings causes a D3D device failure / GPU Crash with 'Standalone Game' mode
- New: 5048577 Application crash due to Error: Render subrect base and size exceed the dimensions of the input Color image

Editor Specific:

- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display

DLSS4Test App Specific:

- New: 5228061 'Auto' Mode in the DLSS4Test app doesn't automatically adjust DLSS resolution, and only inherits the last set DLSS resolution
- New: 5228046 Image corruption occurs when View Rect Offset is adjusted
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- 5046507 Amplified noise on reflections with DLSS-SR Enabled
- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

4.0.0-2.7.3 February 2025

Component	Note
Unreal Engine	5.2 to 5.5
NVIDIA RTX	5.4
UE Platforms	Win64, WinGDK
Streamline SDK	2.7.3
NGX SDK	310.1.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
 - DLSS Super Resolution (DLSS-SR)
 - DLSS Ray Reconstruction (DLSS-RR)
 - Deep Learning Anti Aliasing (DLAA)
 - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
 - RTX Dynamic Vibrance
 - Reflex Low Latency
- D3D12RHI
 - DLSS Frame Generation (DLSS-FG)
 - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

New Features

- DLSS Frame Generation
 - Added support for Multi-Frame Generation
- DLSS Super Resolution
 - Added a new preset, Preset J (Transformer model)
- DLSS Ray Reconstruction
 - General availability
 - Added a new preset, Preset D (Transformer model)

General Changes

Application developer facing:

- The architecture of the family of Streamline plugins has been changed in the 4.0.0 release compared to the 1.x.x releases:
 - Plugins have been renamed
 - Implementation has been split into separate plugins.
 - Some blueprint enums and class names have been changed for UE best practices.
- Throw Blueprint Exception when an invalid BP enum value is passed into BP functions (5.4+)
- Added WinGDK support

Plugin internal:

- Add `r.Streamline.FilterRedundantSetOptionsCalls` and `-slfilter -shofilter` command line options
- NGX DLSS plugin interop at `sllnit` time: only pass in paths to the DLSS-SR/RR NGX binaries when `r.NGX.Enable/r.NGX.Enable.AllowCommandLine` etc would not prohibit doing so on the NGX DLSS plugin.
- Add `r.Streamline.LogFunctions.LogLevel` to customize which functions are logged when `r.Streamline.LogFunctions` is 1
- Log as the UE Error category when Streamline interpose cannot be loaded due to missing `.dll` or similar
- Add `r.Streamline.Load.*` console variables to specify which underlying Streamline binaries are loaded at `sllnit` time
- Add support for specular motion vectors (NVRTX 5.4+)

Bug Fixes

- Fix bug of DLSS-FG engaging unintendedly on notification windows when PIE (This Viewport) by correctly detecting PIE (New Window) where DLSS-FG is supported

Known Issues

Editor Specific:

- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display

DLSS4Test App Specific:

- 5048577 Application crash observed upon toggling different rect offset options
- 5046562 New Reflection artifacts observed with DLSS-SR enabled in triple view
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Upgrading from a project from 1.x.x plugins to 4.x.x plugins

The architecture of the family of Streamline plugins has been changed in the 4.0.0 release compared to the 1.x.x releases:

- Plugins have been renamed
- Implementation has been split into separate plugins
- Some of the blueprint enums and class names have been changed between 1.x.x and 4.x.x to follow overall UE best practises.

Blueprint redirectors have been added to the 4.0.0 release to reduce the friction of upgrading. If those fail for whatever reason the mapping should be straightforward.

Previous 1.x.x plugin architecture

In the *previous* release the 1.x.x. Streamline plugin contained the core of the implementation that connects the engine with the underlying Streamline APIs.

On the developer facing Blueprint side there were a set of Blueprint libraries for generic (`UStreamlineLibrary`) as well as set of feature specific use(such as `UStreamlineLibraryDLSSG` or `UStreamlineLibraryReflex`).

Current 4.x.x plugin architecture

In the *current* 4.x.x releases, the core implementation parts from the 1.x.x Streamline plugin has been renamed to the 4.x.x StreamlineCore plugin. The generic Blueprint library `UStreamlineLibrary` continues to be part of the renamed StreamlineCore plugin.

Note: StreamlineCore is hidden in the editor intentionally since the per feature plugins implicitly enable it.

The feature specific Blueprint libraries from the 1.x.x Streamline plugin however have moved to a set of 4.x.x feature specific plugins such as `StreamlineDLSSG` or `StreamlineReflex`.

Note: StreamlineDeepDVC Blueprint libraries were already separated out in 1.x.x into a separate StreamlineDeepDVC plugin, but now the other features are also separated.

Changes to Blueprint Enum Values

As part of DLSS-MFG support, the enum values of the various enums have changed. This should not impact any well behaved client blueprints that treat enums as opaque identifiers. But any blueprint or C++ code that casts a byte/integer into enum, and thus side-stepping the type safety of Blueprints & C++, will get this reported as a scripting error: `LogScript: Error: Script Msg:`

```
UStreamlineLibraryDLSSG::SetDLSSGMode should not be called with an invalid enum value (252) "EStreamline
```

DLSSGMode MAX"

The fix is to populate the UI element with a list of the enums returned from `UStreamlineLibraryDLSSG::GetSupportedDLSSGModes` and then index into that array via the UI element selection index to get the proper `EStreamlineDLSSGMode` argument for the call to

`UStreamlineLibraryDLSSG::SetDLSSGMode`

Note: `EStreamlineReflexMode` always has had non-contiguous enum *values*, so similar issues might have arisen in the past. A `UStreamlineLibraryReflex::GetSupportedReflexModes` was added in the 4.0.0 release to allow for a similar pattern as for DLSS-FG when populating UE elements with the supported Reflex Low Latency Modes.

Upgrading a 1.x.x project to 4.x.x plugins

The NVIDIA Streamline 1.3.3 Legacy Compatibility plugin (Streamline.uproject) doesn't contain any code and only enables the following plugins to allow projects developed against the 1.x.x family of plugins to continue to work.

- StreamlineCore
- StreamlineDeepDVC
- StreamlineReflex
- StreamlineDLSSG

For the 4.x.x family of plugins, the recommended alternative is to explicitly enable the plugins for each required feature.

Note: Streamline.uproject is hidden in the editor intentionally since it exists purely for existing projects and is not required for new projects

DLSS Generation (DLSS-FG, DLSS-MFG)

The DLSS Multiframe Generation modes have been added as 3X and 4X members to `EStreamlineDLSSGMode`, 2X corresponds to the On mode of the 1.x.x release.

Any existing BP or C++ code that has been using `GetSupportedDLSSGModes` and `UStreamlineLibraryDLSSG::SetDLSSGMode` to implement UI should require no additional effort (other than maybe localization) to have DLSS-MFG be available in a project.

DLSS Super Resolution (DLSS-SR)

A new Preset J project setting has been added to select the DLSS-SR Transformer model

DLSS Ray Reconstruction (DLSS-RR)

DLSS Ray Reconstruction supports stock UE to a certain degree. However the engine modifications in the NVRTX 5.4 branch additional input buffers which will greatly improve the overall quality of DLSS-RR.

A new Preset D project setting has been added to select the DLSS-RR Transformer model

`UDLSSLibrary::IsRayTracingAvailable()` has been added to make implementing UI for DLSS-RR easier

Reflex Low Latency

`UStreamlineLibraryReflex::GetSupportedReflexModes()` has been added to match other BP libraries and make generation UIs easier

1.3.3-SL2.4.15

Release Date: September 2024

- Bug fixes
 - replace viewport tracking related asserts with log messages
 - gracefully handle VR/XR by not asserting
 - actually limit the number of generated swapchain proxies depending on the cvar
- Other Changes
 - Update to Streamline 2.4.15
 - removed UE4 support in the editor (PIE) due to instability related to thread (un)safe accessing of Slate TSharedPtrs
 - enable multi viewport support for DLSS-FG
 - partial 5.5 compatibility
 - various refactorings in the BP libraries to reduce duplicate code
 - add new modes to r.Streamline.DLSSG.AdjustMotionBlurTimeScale to handle auto mode

1.3.2-SL2.4.10

Release Date: June 2024

- Other changes
 - Update to Streamline 2.4.10
 - Support targetting minimum Windows version > Windows 7 (WinGDK platform support)
 - Disable the UE 5.4 workaround when not needed

1.3.1-SL2.4.0

Release Date: May 2024

(Not released on the market place)

- Bug fixes
 - Fix crash on editor startup with D3D11
 - Fix parsing of -slswapchainprovider command line option
 - Fix incorrect public dependencies in StreamlineBlueprint module
- Other changes
 - Support UE 5.4 release

1.3.0-SL2.4.0

Release Date: April 2024

- New features
 - RTX Dynamic Vibrance (DeepDVC)
 - DLSS-FG subrect support
 - Option to enable DLSS-FG dynamic resolution
 - Option to automatically disable DLSS-FG if full screen menus are detected
- Bug fixes
 - Fix assertion with MovieRenderQueue
 - Fix assertion when taking screenshots with F9
 - Fix hudless texture extent
 - Fix viewport resize issue
 - Fix assert when running dumpgpu command
 - Only compile shaders on supported platforms, fixes some console builds
- Other changes
 - Update to Streamline 2.4.0
 - Split PCL and Reflex
 - Removed support for SetInputSampleLatencyMarker
 - SL function logging improvements

1.2.0-SL2.2.1

Release Date: October 2023

- New features

- DLSS-FG auto mode
- Bug fixes
 - Fix startup crash when -RenderOffScreen is used
 - Support running from network paths beginning with \\
- Other changes
 - Update to Streamline 2.2.1
 - Log plugin version on startup
 - Improve error log on DXGI/D3D12 errors from Streamline

1.1.0-SL2.2.0

Release Date: September 2023

- New features
 - Add r.Streamline.InitializePlugin read-only cvar to allow disabling plugin
 - Add FStringStreamlineCoreModule::QueryDLSSGSupport()
- Bug fixes
 - Fix assertion when DXGI_STATUS results are reported by SL SDK. Could happen when app loses focus
- Other changes
 - Support UE 5.3.0 release
 - Disable DLSS-FG when window is not in focus

1.0.7-SL2.2.0

Release date: August 2023

- New features
 - none
- Bug fixes
 - none
- Other changes
 - Update to Streamline 2.2.0
 - Source-level support for UE 5.3.0 preview 1

1.0.6-SL2.1.1

Release date: July 2023

- New features
 - Add r.Streamline.TagUIColorAlphaThreshold cvar for UI extraction pass alpha threshold value
 - Add new blueprint libraries to express various end user recoverable ways why DLSS-FG failed to the game logic
 - Add min supported viewport width and an inaccurate vram estimate to "stat dlssg"
 - Add error handler for DLSS-FG runtime errors
 - Add r.Streamline.DLSSG.CheckStatusPerFrame to check the DLSS-FG status at runtime and assert if it's failing somehow (default = true)
- Bug fixes
 - Fix crash seen during shutdown when SWindow doesn't have a Viewport
 - Disable DLSS-FG if the viewport width is below what DLSS-FG supports at the SL level
 - Fix minor leak in reflex hooks
- Other changes
 - Update to Streamline 2.1.1
 - Disable DLSS-FG on scene captures

1.0.5

- there was no version 1.0.5 plugin release

1.0.4-SL2.1.0

Release date: June 2023

Initial public plugin release